Question ID: 752055d1

Question

A scientist initially measures $12{,}000$ bacteria in a growth medium. 4 hours later, the scientist measures $24{,}000$ bacteria. Assuming exponential growth, the formula $P = C(2)^{rt}$ gives the number of bacteria in the growth medium, where r and C are constants and P is the number of bacteria t hours after the initial measurement. What is the value of r?

Answer

A. $\frac{1}{12{,}000}$

B. $\frac{1}{4}$

C. 4

D. $12{,}000$